Post-Quantum Cryptography Migration

Part 5 - Cryptographic Foundations: HSMs, KMS & Key Management

Compiled by the Spaced Research Ledger

Last updated: 2026-08-29

Hardware security modules, key managers, and the standards around them are where organizations keep their most sensitive cryptographic keys. This layer has to adopt post-quantum algorithms before much else can. A certificate authority or a payments network can only move to a new algorithm once the hardware beneath it supports that algorithm and auditors accept the result.

This report covers the layer in five sections. Section 1 covers the HSM vendors, Section 2 the key-management interoperability standard, and Section 3 the government validation pipeline that gates procurement. Section 4 covers the adjacent quantum-hardware market, and Section 5 the problem of keys and archives that must survive the transition.

The most important shift this summer is that validated post-quantum hardware finally exists. As late as December 2025, no HSM held a FIPS 140-3 Level 3 validation that included the new algorithms. By August 2026 several had crossed that line, though two vendors' competing claims to being first remain unreconciled, and Section 3 records the dispute.

Products and protocols are moving with the validations. Thales launched its Luna 8 line built for post-quantum performance. KMIP, the protocol key managers use to interoperate, is nearing a 3.0 release that adds the new algorithms directly, with a three-vendor interoperability test already passed. Cloud key services are moving too. Quantum key distribution and quantum random numbers remain a separate, adjacent market: real hardware, heavy marketing, and no substitute for the algorithm migration itself.

1. HSM Vendors

A hardware security module is a tamper-resistant device that generates and guards cryptographic keys so they never leave protected hardware. Banks, governments, and cloud providers anchor their trust chains in these devices. Post-quantum migration starts here: an organization cannot issue quantum-safe certificates or signatures until its HSMs can hold and use the new keys.

Recent Developments

Validated post-quantum HSMs arrived this summer. Thales TCT's Luna T-Series achieved FIPS 140-3 Level 3 validation in August 2026 [1]. It is described as the first post-quantum HSM made in the United States to reach that level [1]. It is also called the first of any origin to include the full NSA CNSA 2.0 suite in one validation [1].

Entrust's nShield 5s PCIe modules hold the same validation level with native, hardware-accelerated ML-DSA, ML-KEM, and SLH-DSA [2]. Entrust publishes concrete signing performance and support for eighteen SLH-DSA variants in its datasheets [3].

New hardware followed the validations. Thales launched the Luna 8 Network HSM in August 2026, three models with a custom cryptographic processor claiming up to 100x faster post-quantum performance [4]. The upgradeable architecture is now under FIPS and Common Criteria evaluation [4]. Utimaco's Quantum Protect algorithms passed NIST validation with specific certificate references [5].

Current State

Thales built to this point in stages. It shipped a FIPS-compliant HSM generating quantum-enhanced keys in 2021, added pre-standardization algorithm support from 2023, and has sat in the NIST migration project since 2022 [6]. Luna firmware 7.9.0 folded ML-KEM and ML-DSA directly into the firmware, ending the need for separate add-on modules [7], with LMS support arriving separately [8]. Its network encryptors support all three NIST algorithms and integrate with quantum key distribution links [9].

Entrust's line is fully validated at the algorithm level. Firmware 13.8.0 provides native ML-DSA, ML-KEM, and SLH-DSA [10] and holds NIST algorithm validation for them [10]. An industry capability matrix, updated by the vendor in March 2025, still showed the hash-based schemes as unsupported at that point [11].

Utimaco packaged its migration as a field upgrade. Quantum Protect, launched April 2025, activates ML-KEM, ML-DSA, LMS, and XMSS on existing hardware, with a patented state-management approach for the stateful schemes and a free simulator [12]. NXP is a named customer, using it to sign firmware updates with ML-DSA [13]. The Se-Series hardware gained a FIPS 140-3 Level 3 certificate in May 2026 [14], and SLH-DSA sits on the roadmap [15].

The cloud-facing vendors trail. Marvell's LiquidSecurity hardware behind Azure's HSM services was designed field-upgradable in 2022 without claiming any specific algorithm [16]. A July 2026 analysis found ML-KEM and ML-DSA still unsupported there [17]. AWS CloudHSM long treated post-quantum keys as opaque storage [17], though AWS's own migration page now lists ML-DSA generation and signing in preview on that hardware [18].

Yubico remains at the prototype stage: its post-quantum hardware key demonstration was explicitly not a shipment announcement, since the algorithms do not fit today's keys [19]. Its protocol work adds forward-looking plumbing while noting no standard post-quantum attestation formats exist [20], and its recent FIPS validations cover classical cryptography [21].

Unresolved Conflicts

Utimaco's validation status is inconsistent across its own materials. One product page calls the Se-Series FIPS 140-2 certified, another says FIPS 140-3 validated, and a third lists the newer series as in progress [13]. A later press release claims a definitive Level 3 certification with a hedged certificate number without resolving the inconsistency [14]. A NIST certificate lookup would settle it. Separately, one Utimaco source says SLH-DSA is already supported via the package [5], contradicting the roadmap status reconfirmed elsewhere [15].

2. Key Management Interop

Key managers are the systems that distribute and rotate an organization's encryption keys. KMIP is the shared protocol that lets key-management products from different vendors work together. If the protocol cannot describe a post-quantum key, no compliant product can manage one. That makes this quiet standards work a gating item for enterprise migration.

Recent Developments

KMIP 3.0 entered its final public stretch. OASIS opened comment periods on the specification and profiles ending August 13, 2026 [22], and on the usage guide ending two days later [23]. The usage guide carries a dedicated post-quantum section [24]. The profiles define Quantum Safe client and server conformance levels requiring specific algorithms, hybrid KEMs, and quantum-safe TLS [25]. Cryptsoft's C and Java SDKs now include the post-quantum algorithms by default [26].

Current State

The ratified standard still predates the algorithms. KMIP 2.1, approved in 2020, defines no post-quantum object types, while 3.0 remains a committee draft [27]. The draft's additions are concrete: normative references to the three FIPS standards, a KEM enumeration, and new Encapsulate and Decapsulate operations [27]. A client can then run key encapsulation through the key manager itself [27].

One useful feature already shipped. A Quantum Safe attribute, present since version 2.0, lets deployed key managers tag and query quantum-vulnerable keys today [27]. That makes them working inventory tools ahead of the migration [27].

Implementations are ahead of ratification. A March 2025 interoperability test ran three vendors through the full 1,452-test post-quantum set, exchanging over a million messages and verifying hybrid TLS [28]. Entrust's KMIP vault creates quantum-safe objects and performs the new operations [29], and Cryptsoft's server SDK covers KMIP 3.0 with the NIST algorithms and hybrid TLS [30].

Buyers should ask two separate questions: which KMIP versions a product speaks, and what protects the session itself [31]. KMIP mandates mutual TLS, whose quantum-safety depends on the server's own TLS stack [31]. Thales's CipherTrust release notes illustrate the distinction, documenting post-quantum transport protection without confirming the new managed-key object types [32].

3. FIPS 140-3 Validation Status

FIPS 140-3 is the U.S. and Canadian government standard for certifying that a cryptographic product does what it claims. Federal agencies and many regulated industries may only buy validated products. Until post-quantum algorithms appear inside these certificates, large buyers cannot deploy them. That makes the validation queue a real bottleneck for the whole migration.

Recent Developments

The dam broke in August 2026. Crypto4A's QASM module achieved FIPS 140-3 Level 3 validation supporting the full NIST post-quantum suite [33], and DigiCert promptly integrated it into its signing platform [34]. Sansec's HSM received certificate #5503, the first Chinese module supporting the NIST algorithms at that level [35].

The program's rulebook is being updated in step. August guidance added self-test requirements for the three FIPS standards and permitted internal algorithm self-testing [36], clarified key-generation checks [36], and added a new section on hybrid modules [37]. The in-process list shows the queue behind the pioneers, with SafeLogic in review and other post-quantum libraries cycling through resubmission [38]. PQShield's core library sits in lab comment resolution [39], and fresh algorithm certificates keep landing [40].

Current State

Validation runs as a two-stage pipeline: algorithm validation first, then the full module [41]. Modules then progress through published in-process phases before a certificate issues [42]. AWS-LC illustrates the cadence, with certificates in 2023 and 2024 and a third version entering the queue in December 2024 as the first open-source module with ML-KEM in scope [43]. SafeLogic's provider completed algorithm validation and entered the module queue in May 2026, expecting full validation by year end [44].

Capacity is the structural problem. The program has validated over 5,000 modules in its history [45]. Every remaining FIPS 140-2 certificate moves to the historical list in September 2026, compounding demand exactly as post-quantum submissions surge [45]. FIPS 140-3 went from 3 percent of certificates in 2023 to 45 percent in early 2024 [46]. NIST's answer is automation: a draft practice guide aims to close the capacity gap, alongside interim-validation options already in effect [47].

Unresolved Conflicts

Two vendors claim the same first. Thales TCT's August validation is described as the first FIPS 140-3 Level 3 for a post-quantum HSM made in the United States, with the full CNSA 2.0 suite [1]. Crypto4A's is described as the first quantum-safe HSM at that level outright [33]. The claims are worded to coexist, but coverage treats each as the milestone. Certificate dates on the NIST registry would settle the order.

Both postdate the December 2025 assessment that no vendor had yet combined Level 3 validation with post-quantum support [48].

4. QRNG & QKD Adjacency

Quantum key distribution and quantum random number generation are hardware technologies that use quantum physics itself, unlike post-quantum cryptography, which is ordinary software math. QKD ships encryption keys over dedicated optical links. QRNG produces high-quality randomness. Both are real and growing, but neither replaces the algorithm migration, so this section covers them as neighbors rather than substitutes.

Recent Developments

The QKD demonstrations keep scaling. Researchers ran a side-channel-secure protocol over 200 kilometers of fiber with measured key rates [49], and an Italian team generated keys across an 18-kilometer hybrid air-and-fiber link [50]. New York State lit the first U.S. free-space quantum link, 13 miles between two labs on a 161-mile network [51]. Chinese researchers demonstrated device-independent QKD over 11 kilometers using entangled atoms [52].

India's defense research agency completed military field trials of an indigenous fiber system [53].

The commercial ecosystem is organizing. ID Quantique, now under IonQ, released a product multiplexing quantum keys with data traffic on existing metropolitan fiber [54]. A European consortium launched a four-year project to certify industrial continuous-variable QKD [55]. SK Telecom won European funding for a next-generation system [56], and ESA's QKD satellite is slated for 2026 [57].

On the randomness side, labs pushed generation rates to tens of gigabits per second [58][59], a self-testing chip was published [60], and a rack-mounted certified system launched [61]. NIST demonstrated certifiably random bits from a Bell test [62]. One vendor submitted an entropy module for NIST validation [63], and a chipmaker won an innovation award for the smallest QRNG [64]. One project is integrating QKD and QRNG onto a single photonic chip [65].

Current State

The technologies are neighbors, not substitutes. Post-quantum cryptography is new math on classical hardware, QKD is photons on dedicated optics, and QRNG is an entropy source that feeds either [66]. Industry guidance says to migrate algorithms first and evaluate QKD selectively for the most sensitive links [66]. The consolidation trend shows layering rather than competition, with China's carrier deploying hybrid QKD-plus-PQC systems in production [66].

Deployment is regional and government-led. China linked Beijing to South Africa on its Micius-derived network [67], Singapore's national program is trialing Southeast Asia's first quantum-secure network [68], and IonQ plans a satellite QKD constellation [67]. Patent filing patterns suggest movement toward carrier-grade integration [69]. Germany's telecom demonstrated quantum teleportation over live commercial fiber [57].

The market numbers deserve skepticism. QKD estimates put 2026 at about $1.6 billion with aggressive growth projections, with the sources themselves cautioning about promotional tendencies [68]. QRNG forecasting has been worse: a 2021 report projected $7.2 billion by 2026, while actual 2026 figures cluster around half a billion [70]. One commentary describes a gold-rush phase with over 120 companies and marketing filling the gap where the physics offers little differentiation [71]. Real engineering continues underneath, including Toshiba's fully chip-integrated QRNG [72], national investment programs across dozens of countries [73].

5. Key Escrow, Archive & Re-encryption

Encrypted data that must stay readable for decades, such as backups, archives, and escrowed keys, is the most exposed part of any organization to harvest-now-decrypt-later attacks. An adversary can copy ciphertext today and wait for a quantum computer. The defense is re-wrapping stored keys under quantum-safe algorithms, which turns out to be far cheaper than re-encrypting the data itself.

Recent Developments

The cloud providers moved on key import and transport. Google Cloud KMS previewed quantum-safe key import, wrapping imported key material with post-quantum encryption in transit [74]. Veeam's backup platform added hybrid post-quantum options on its key transport paths, aimed squarely at harvest-now-decrypt-later exposure for long-lived backups [75]. NIST finalized the update to its crypto-agility white paper [76], and the Trusted Computing Group's new guidance defines quantum-ready hardware categories for key custody [77]. Thales's Luna 8 targets the same custody market with its upgradeable architecture [78].

Current State

The central insight is that re-wrapping beats re-encrypting. Database migrations replace only the key-wrapping layer, moving the key-encryption key to ML-KEM while the bulk data stays under its existing symmetric encryption [79]. The procedure is mechanical: generate a new post-quantum wrapping key, re-wrap the data keys, retire the old key, never touch the data files [79]. Academic constructions formalize the same pattern [80], and Microsoft's database encryption already versions its ciphertext for exactly this kind of graceful transition [81]. Research prototypes go further, layering QKD and proxy re-encryption for cloud storage [82].

Government guidance frames the deadline. NIST's transition report starts deprecating quantum-vulnerable public-key algorithms after 2030, with broader disallowance after 2035, defining the window for re-wrapping archives [83]. Its crypto-agility paper treats agility as an architectural property, not a one-time migration [83], and standing advice is to rotate or re-encrypt before algorithms break, not after [84]. The key-management guidelines are being revised to incorporate the new algorithms [85], and the KEM recommendation adds operational rules, including that an ephemeral private key be used exactly once [86]. The transition framing designates ML-KEM as the successor to both classical key-establishment families [87].

The cloud key services are the furthest along in practice. AWS KMS combines hybrid post-quantum TLS to its endpoints with ML-DSA signing operations [88], built on the first open-source FIPS module with ML-KEM [89]. It swapped its earlier pre-standard Kyber support for ML-KEM across services [90]. It is extending ML-DSA into its certificate services for quantum-resistant roots of trust [89], and uses key combiners merging classical and post-quantum KEMs for wrapping [88].

Its public certificates remain classical while browser support lags [91]. Google's KMS supports the full algorithm set for generation, encapsulation, and signatures [88].

The hardware custody chain moves slowest. HSM firmware gained the algorithms in late 2024, but critical-infrastructure deployments run on decade-plus replacement cycles [92], and the validation gap only closed this August, as Section 3 records [48]. Escrow is the research frontier. Prototype post-quantum escrow systems exist for lawful-access scenarios [93], motivated by the quantum vulnerability of the classical escrow protocols [94]. At least one IETF draft already mandates ML-DSA for escrow operations [95].

About This Report

The Spaced Research Ledger is an automated system that gathers claims from live public sources, checks each one against its origin, and records every disagreement instead of merging it. Every stage runs on a different AI model, drawn from four to seven systems across competing labs, so no single model both finds a claim and judges it. The Spaced Research Ledger is designed and maintained by Ridley DiSiena.

Sources

1. intelligencecommunitynews.com — thales tcts post quantum hsm achieves fips 140 3 level 3 - https://intelligencecommunitynews.com/thales-tcts-post-quantum-hsm-achieves-fips-140-3-level-3/ - as of 2026-07-29
2. nShield 5s PCIe HSM | Entrust - https://www.entrust.com/products/hsm/nshield-5s - as of 2026-08-27
3. nShield 5s HSM datasheet - https://www.entrust.com/sites/default/files/documentation/datasheets/dps-entrust-nshield-5s-ds.pdf - as of 2026-06-08
4. Introducing Luna 8: Our Fastest HSM Ever, Built for the Post-Quantum Era - https://data-protection-updates.gemalto.com/2026/08/04/introducing-luna-8-our-fastest-hsm-ever-built-for-the-post-quantum-era/ - as of 2026-08-04
5. Utimacos Nist Validated PQC Algorithms Setting Standard - https://utimaco.com/news/blog-posts/utimacos-nist-validated-pqc-algorithms-setting-standard-future-data-security - as of 2026-07-31
6. thalestct.com — hsm fips140 3 validation - https://www.thalestct.com/hsm-fips140-3-validation/ - as of 2026-08-04
7. cpl.thalesgroup.com — luna hsm pqc quantum safe encryption - https://cpl.thalesgroup.com/blog/encryption/luna-hsm-pqc-quantum-safe-encryption - as of 2025-07-29
8. thalesdocs.com — post quantum algorithms - https://thalesdocs.com/gphsm/luna/7/docs/network/Content/sdk/extensions/pqc/post_quantum_algorithms.htm - as of 2026-04-07
9. cpl.thalesgroup.com — post quantum crypto agility - https://cpl.thalesgroup.com/encryption/post-quantum-crypto-agility
10. Entrust nShield HSMs Post-Quantum Cryptography ... - https://www.entrust.com/company/newsroom/entrust-nshield-hsms-achieve-validation-from-nist-cryptographic-algorithm-validation-program - as of 2025-09-10
11. pkic.org — pqccm - https://pkic.org/wg/pqc/pqccm/ - as of 2025-03-01

12. Utimaco launches Quantum Protect solution and free PQC simulator - https://utimaco.com/news/press-releases/utimaco-launches-quantum-protect-solution-and-free-pqc-simulator - as of 2025-04-03
13. utimaco.com — utrust general purpose hsm - https://utimaco.com/data-protection/gp-hsm/utrust-general-purpose-hsm
14. Utimaco Vendor Offer FIPS 140-3 Certification Hardware - https://utimaco.com/news/press-releases/utimaco-only-vendor-offer-full-fips-140-3-certification-its-hardware-security - as of 2026-05-12
15. The 2029 Post-Quantum Deadline: Can Your HSMs Make the Migration? - https://utimaco.com/news/blog-posts/post-quantum-hsm-migration-2029 - as of 2026-04-22
16. marvell.com — marvell liquidSecurity2 hardware security multi cloud era - https://www.marvell.com/company/newsroom/marvell-liquidSecurity2-hardware-security-multi-cloud-era.html - as of 2022-09-14
17. quantumsecuritydefence.com — cloud kms pqc readiness aws azure gcp - https://quantumsecuritydefence.com/insights/cloud-kms-pqc-readiness-aws-azure-gcp/ - as of 2026-07-13
18. aws.amazon.com — migrating to post quantum cryptography - https://aws.amazon.com/security/post-quantum-cryptography/migrating-to-post-quantum-cryptography/
19. yubico.com — future proofing authentication a look at the future of post quantum cryptography - https://www.yubico.com/blog/future-proofing-authentication-a-look-at-the-future-of-post-quantum-cryptography/ - as of 2026-01-14
20. docs.yubico.com — yk5 apps fido - https://docs.yubico.com/hardware/yubikey/yk-tech-manual/yk5-apps-fido.html
21. yubico.com — leading yubico forward q2 reflections and securing the ai frontier - https://www.yubico.com/blog/leading-yubico-forward-q2-reflections-and-securing-the-ai-frontier/
22. Invitation to comment on KMIP Specification v3.0 and ... - https://www.oasis-open.org/2026/07/14/invitation-to-comment-on-kmip-specification-v3-0-and-kmip-profiles-v3-0-ends-13-august-2026/ - as of 2026-07-14
23. Invitation to comment on KMIP Usage Guide v3.0 - https://www.oasis-open.org/2026/07/16/invitation-to-comment-on-kmip-usage-guide-v3-0-ends-15-august-2026/ - as of 2026-07-16
24. Key Management Interoperability Protocol Usage Guide Version 3.0 - https://docs.oasis-open.org/kmip/kmip-ug/v3.0/cnd01/kmip-ug-v3.0-cnd01.pdf - as of 2026-06-18
25. Key Management Interoperability Protocol Profiles Version 3.0 - https://docs.oasis-open.org/kmip/kmip-profiles/v3.0/kmip-profiles-v3.0.pdf - as of 2026-07-13
26. Post-quantum, by default: PQC in every Cryptsoft SDK - https://cryptsoft.com/blog-pqc-by-default.html - as of 2026-07-01
27. encryptionconsulting.com — kmip - https://www.encryptionconsulting.com/education-center/kmip/
28. Cryptsoft KMIP Server - KMIP Interop - https://kmip-interop.org/background.html - as of 2025-03-07

29. Post Quantum Support in the Cryptographic Security Platform Vault for KMIP - https://docs.hytrust.com/CryptographicSecurityPlatformVault/Online/Content/Books/Admin-Guide/KIMP-Vault/PQ-in-KMIP-Vault.html - as of 2026-03-17
30. KMIP C Server SDK - Cryptsoft - https://cryptsoft.com/product-kmipc_server.html - as of 2026-04-27
31. qcecuring.com — kmip protocol explained - https://www.qcecuring.com/blog/kmip-protocol-explained - as of 2026-05-11
32. data-protection-updates.gemalto.com — ciphertrust manager 2 - https://data-protection-updates.gemalto.com/category/ciphertrust-manager-2/ - as of 2026-03-30
33. Crypto4A Achieves World-First FIPS 140-3 Level 3 Validation for Quantum-Safe HSM Module - https://quantumcomputingreport.com/crypto4a-achieves-world-first-fips-140-3-level-3-validation-for-quantum-safe-hsm-module/ - as of 2026-08-20
34. Crypto4A Sets Global Benchmark with FIPS 140-3 Level 3 Validation for Quantum-Safe HSM Technology - https://www.prnewswire.com/news-releases/canadian-cybersecurity-company-crypto4a-sets-global-benchmark-with-fips-140-3-level-3-validation-for-next-generation-quantum-safe-hsm-technology-302855762.html - as of 2026-08-21
35. 三未信安取得 FIPS 140-3 Level 3 认证，全球化布局再落关键一子 - https://www.stcn.com/article/detail/4138697.html - as of 2026-08-27
36. FIPS 140-3 IG and RFG Announcements - https://csrc.nist.gov/projects/cryptographic-module-validation-program/fips-140-3-ig-announcements - as of 2026-08-14
37. Cryptographic Module Validation Program CMVP - FIPS 140-3 IG and RFG Announcements - https://csrc.nist.gov/projects/cryptographic-module-validation-program/fips-140-3-ig-announcements - as of 2026-08-19
38. Cryptographic Module Validation Program CMVP - Modules In Process List - https://csrc.nist.gov/projects/cryptographic-module-validation-program/modules-in-process/modules-in-process-list - as of 2026-08-14
39. Cryptographic Module Validation Program CMVP - Modules In Process List - https://csrc.nist.gov/Projects/cryptographic-module-validation-program/modules-in-process/Modules-In-Process-List - as of 2026-08-26
40. PQCryptoLib-Core 3.5.1 achieves CAVP certification - https://semiwiki.com/forum/threads/pqcryptolib-core-3-5-1-achieves-cavp-certification.25718/ - as of 2026-08-17
41. businesswire.com — en - https://www.businesswire.com/news/home/20250910020412/en
42. support.apple.com — web - https://support.apple.com/guide/certifications/cryptographic-module-validation-status-apc33ea4bd77/web - as of 2022-12-01
43. aws.amazon.com — aws lc fips 3 0 first cryptographic library to include ml kem in fips 140 3 validation - https://aws.amazon.com/blogs/security/aws-lc-fips-3-0-first-cryptographic-library-to-include-ml-kem-in-fips-140-3-validation/ - as of 2024-12-10
44. safelogic.com — cryptocomply 140 3 fips provider with pqc enters cmvp queue - https://www.safelogic.com/blog/cryptocomply-140-3-fips-provider-with-pqc-enters-cmvp-queue - as of 2026-05-28

45. csrc.nist.gov — cryptographic module validation program - https://csrc.nist.gov/projects/cryptographic-module-validation-program
46. safelogic.com — fips 140 3 - https://www.safelogic.com/fips-140-3 - as of 2023-06-14
47. industrialcyber.co — nist advances cmvp modernization to close gap between cryptographic innovation and validation capacity - https://industrialcyber.co/nist/nist-advances-cmvp-modernization-to-close-gap-between-cryptographic-innovation-and-validation-capacity/ - as of 2026-04-17
48. PQC HSM: How Hardware Security Modules Fit In PQC Migration | QNu Labs - https://www.qnulabs.com/blog/pqc-hsm - as of 2025-12-01
49. Proof-of-principle experimental demonstration of side-channel-secure quantum key distribution over 200 km - https://www.nature.com/articles/s41534-026-01346-4 - as of 2026-08-11
50. Intermodal quantum key distribution over an 18 km free-space channel with adaptive optics - https://arxiv.org/abs/2602.16680 - as of 2026-08-26
51. Governor Hochul Announces Milestone in Expansion of New York State's Quantum Communications Network - https://www.governor.ny.gov/news/governor-hochul-announces-milestone-expansion-new-york-states-quantum-communications-network - as of 2026-08-21
52. A hack-proof internet? Quantum encryption could be the key - https://www.science.org/content/article/hack-proof-internet-quantum-encryption-could-be-key - as of 2026-08-21
53. DRDO Successfully Completes Military Field Trials of Indigenous Fiber-Based Quantum Key Distribution System - https://www.thedefensenews.com/DRDO-Successfully-Completes-Military-Field-Trials-of-Indigenous-Fiber-Based-Quantum-Key-Distribution-System/ - as of 2026-07-15
54. ID Quantique and IonQ Introduce Clavis XG Multiplex to Deploy Quantum Key Distribution Over Metro Networks - https://quantumcomputingreport.com/id-quantique-and-ionq-introduce-clavis-xg-multiplex-to-deploy-quantum-key-distribution-over-metro-networks/ - as of 2026-06-18
55. European Consortium QUARTERNEXT Launches to Advance Certifiable Quantum-Key Distribution Systems - https://quantumcomputingreport.com/european-consortium-quarternext-launches-to-advance-certifiable-quantum-key-distribution-systems/ - as of 2026-07-09
56. SK Telecom Develops Next-Generation Quantum Cryptography Technology with EU Research Funding - https://news.sktelecom.com/en/3147 - as of 2026-08-29
57. Quantum Key Distribution (QKD) Companies 2026 | Directory - https://entangledfuture.com/qkd/ - as of 2026-08-13
58. Highly integrated broadband entropy source for quantum random number generators based on vacuum fluctuations - https://arxiv.org/html/2504.18795v1 - as of 2026-08-24
59. 40Gbps Tri-type Quantum Random Number Generator - https://arxiv.org/html/2506.05627v1 - as of 2026-08-24
60. Researchers unveil self-testing quantum random chip - https://datacenternews.asia/story/researchers-unveil-self-testing-quantum-random-chip - as of 2026-06-19
61. Fraunhofer IPMS Launches 19-Inch Rack QRNG System - https://quantumzeitgeist.com/fraunhofer-ipms-19-inch-rack-qrng/ - as of 2026-06-22

62. New Quantum Method Generates Really Random Numbers - https://www.afcea.org/signal-media/technology/new-quantum-method-generates-really-random-numbers - as of 2026-08-26
63. Quantum eMotion’s Quantum Entropy Source Submitted for NIST Validation - https://thequantuminsider.com/2026/08/27/quantum-emotion-submits-quantum-entropy-module-for-nist-review/ - as of 2026-08-27
64. Elmos wins the 2026 German Innovation Award for the world’s smallest quantum random number generator (QRNG) - https://www.elmos.com/english/about-elmos/newsroom/press-releases/news/elmos-wins-the-2026-german-innovation-award-for-the-worlds-smallest-quantum-random-number-generator-qrng.html - as of 2026-04-20
65. ThinkQuantum Initiates Photonic Chip Miniaturization Program for Terrestrial and Space Cryptography - https://quantumcomputingreport.com/thinkquantum-initiates-photonic-chip-miniaturization-program-for-terrestrial-and-space-cryptography/ - as of 2026-06-09
66. thequantuminsider.com — 25 companies building the quantum cryptography communications markets - https://thequantuminsider.com/2026/03/25/25-companies-building-the-quantum-cryptography-communications-markets/ - as of 2026-06-17
67. coherentmarketinsights.com — quantum key distribution market 5971 - https://www.coherentmarketinsights.com/market-insight/quantum-key-distribution-market-5971 - as of 2026-05-13
68. sptel.com — quantum key distribution - https://sptel.com/quantum-key-distribution/ - as of 2026-05-13
69. patsnap.com — quantum key distribution networks 2026 patsnap eureka - https://www.patsnap.com/resources/blog/rd-blog/quantum-key-distribution-networks-2026-patsnap-eureka/ - as of 2026-04-29
70. globenewswire.com — New IQT Research Report Quantum Random Number Generators will become a 7.2 Billion Market by 2026 - https://www.globenewswire.com/news-release/2021/01/26/2164365/0/en/New-IQT-Research-Report-Quantum-Random-Number-Generators-will-become-a-7-2-Billion-Market-by-2026.html - as of 2021-01-26
71. postquantum.com — qrng buyers guide - https://postquantum.com/post-quantum/qrng-buyers-guide/ - as of 2026-05-29
72. marketsandmarkets.com — quantum random number generator market 208446014 - https://www.marketsandmarkets.com/Market-Reports/quantum-random-number-generator-market-208446014.html
73. marketgrowthreports.com — quantum random number generator qrng chip market 113457 - https://www.marketgrowthreports.com/market-reports/quantum-random-number-generator-qrng-chip-market-113457 - as of 2026-01-13
74. Announcing quantum-safe key import in Cloud KMS - https://cloud.google.com/blog/products/identity-security/announcing-quantum-safe-key-import-in-cloud-kms - as of 2026-08-20
75. Post-Quantum Cryptography and Backup Readiness - https://www.veeam.com/blog/post-quantum-cryptography-backup.html - as of 2026-06-18

76. CSWP 39upd1, Considerations for Achieving Crypto Agility: Strategies and Practices - https://csrc.nist.gov/pubs/cswp/39/upd1/considerations-for-achieving-crypto-agility/final - as of 2026-06-29
77. New Guidance Helps Businesses Verify Quantum-Safe Hardware Claims - https://www.infosecurity-magazine.com/news/guidance-verify-quantum-safe/ - as of 2026-08-24
78. Thales launches Luna 8 hardware security module in India, targeting post-quantum cryptography readiness for enterprises and government - https://www.91mobiles.com/updates/thales-launches-luna-8-hardware-security-module-in-india/ - as of 2026-08-29
79. PQC at the Application Layer: Database Encryption Migration - https://quantumsecuritydefence.com/insights/pqc-application-layer-database-encryption-migration/ - as of 2026-07-06
80. Seamless Post-Quantum Transition - https://www.scitepress.org/Papers/2025/136410/136410.pdf
81. NIST Defines Crypto-Agility As Key To PQC Resilience Says Microsoft - https://quantumzeitgeist.com/microsoft-nist-defines-crypto-agility-key/
82. QuCloud : Enhancing cloud storage security by combining quantum key distribution, post-quantum cryptography, and custom proxy re-encryption - https://www.sciencedirect.com/science/article/abs/pii/S2214212626000797 - as of 2026-06-01
83. Cryptographic Agility: Your First Line of Defense Against Quantum Risk - https://www.encryptionconsulting.com/cryptographic-agility-defense-against-quantum-risk/ - as of 2025-12-01
84. Cryptographic Agility: The Key to Quantum Readiness - Palo Alto Networks - https://www.paloaltonetworks.com/cyberpedia/what-is-cryptographic-agility
85. NIST Special Publication (SP) 800-57 Rev. 6 (Draft), Recommendation for Key Management: Part 1 – General - https://csrc.nist.gov/pubs/sp/800/57/pt1/r6/ipd - as of 2025-12-05
86. NIST SP 800-227: Recommendations for Key-Encapsulation Mechanisms - https://csrc.nist.gov/pubs/sp/800/227/final - as of 2025-09-01
87. NIST IR 8547 initial public draft, Transition to Post-Quantum Cryptography Standards - https://nvlpubs.nist.gov/nistpubs/ir/2024/NIST.IR.8547.ipd.pdf
88. Is the Public Cloud Ready for Post-Quantum Cryptography? | AWS in Plain English - https://aws.plainenglish.io/is-the-public-cloud-ready-for-post-quantum-cryptography-a035d82fd0e2?gi=18fdfaa9daec
89. Post-Quantum Cryptography - Amazon Web Services - https://aws.amazon.com/security/post-quantum-cryptography/
90. AWS introduces post-quantum security with ML-KEM for TLS connections - Techzine Global - https://www.techzine.eu/news/security/130431/aws-introduces-post-quantum-security-with-ml-kem-for-tls-connections/ - as of 2025-04-01
91. AWS KMS Post-Quantum: ML-KEM & ML-DSA in Production 2026 - https://www.factualminds.com/blog/aws-kms-post-quantum-cryptography-ml-kem-ml-dsa/
92. Securing Cryptography in the Age of Quantum Computing and AI: Threats, Implementations, and Strategic Response - https://arxiv.org/pdf/2603.06969 - as of 2024-12-01
93. Post-quantum Key Escrow for Supervised Secret Data Sharing on Consortium Blockchain | SpringerLink - https://link.springer.com/chapter/10.1007/978-3-030-94029-4_12

94. Enhanced post-quantum key escrow system for supervised data conflict of interest based on consortium blockchain - PMC - https://pmc.ncbi.nlm.nih.gov/articles/PMC10239057/
95. draft-ailex-vap-legal-ai-provenance-03 - https://datatracker.ietf.org/doc/draft-ailex-vap-legal-ai-provenance-03/ - as of 2026-03-01